

Mode locking with cross-phase and self-phase modulation

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Cross-phase and self-phase modulation are used for self-sustained mode locking of a high-power neodymium glass fiber laser. Stable pulses with a FWHM as short as 70 fs and pulse energies of as much as 1 nJ are generated at a wavelength of 1.064 μm .

Ultrashort-pulse generation in lasers generally relies on the simultaneous presence of both amplitude modulation (AM) and frequency modulation (FM) at the cavity round-trip frequency,¹ where most conveniently these effects are induced passively. In single-mode glass fibers passive FM may be obtained by using self-phase modulation in the presence of overall negative group-velocity dispersion² (GVD). On the other hand, passive AM in fibers may be introduced by an all-optical switch.^{3,4} AM is required for the generation of pedestal-free pulses, since any low-intensity nonlinear contributions to the frequency chirp are continuously filtered out by the switching action,⁵ and under optimum conditions passive AM may also lead to self-starting passive mode locking.⁶⁻⁸

We show here that in a highly nonlinear laser cavity passive AM may be obtained by nonlinear polarization evolution owing to the combined action of self-phase modulation and cross-phase modulation. In this case, nonreciprocal pulse shaping is obtained in a slightly linearly birefringent fiber inserted into a single-polarization Fabry-Perot cavity. Owing to the high pulse intensities required to demonstrate the effect, we use a single-mode rare-earth-doped fiber⁹ as the active medium, which provides a useful combination of large gain and nonlinearities. Our results may also be related to recent research on high-power bulk lasers,^{10,11} which can exhibit self-starting mode locking in a single cavity.

The fabrication of fibers with various degrees of birefringence¹² is now well established, and linear¹³ and nonlinear¹⁴ propagation in such media are well understood. In our experiments we deal with nontwisted slightly linearly birefringent fibers, and therefore we initially limit our discussion to this simple case. Consider the simplified Fabry-Perot laser cavity in Fig. 1(a). Here the x and y axes are assumed to be aligned with the linearly polarized eigenmodes of the fiber, and a polarizer is inserted into the cavity at an angle α with respect to the x axis. A signal of amplitude A that passes the polarizer from the left and is coupled into the fiber will then be linearly polarized and excite the x and y axes with amplitudes $A_x = A \cos \alpha$ and $A_y = A \sin \alpha$, respectively. Both waves then suffer differential linear and nonlinear phase delays and nonlinear coupling, which can be calculated from solutions of the appropriate coupled-wave equa-

tions.¹⁵ In the rotating-wave approximation (RWA), which applies for typical round-trip linear phase delays between A_x and $A_y \geq 3\pi$, nonlinear coupling may be neglected,¹⁶ and the total phase delays along the x and y axes are simply given by adding the linear, self-phase, and cross-phase modulation terms,

$$\Phi_x = \{\beta_x + kn_2[|A_x|^2 + (2/3)|A_y|^2]\}2L, \quad (1a)$$

$$\Phi_y = \{\beta_y + kn_2[|A_y|^2 + (2/3)|A_x|^2]\}2L, \quad (1b)$$

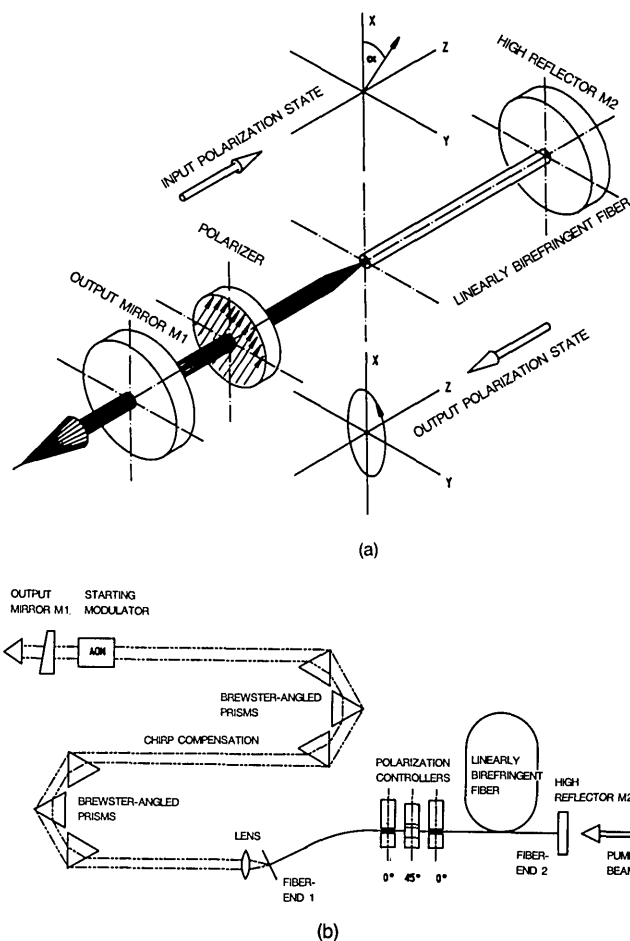


Fig. 1. (a) Operation principle of a nonlinear Fabry-Perot cavity and (b) its experimental verification. AOM, acousto-optic modulator.

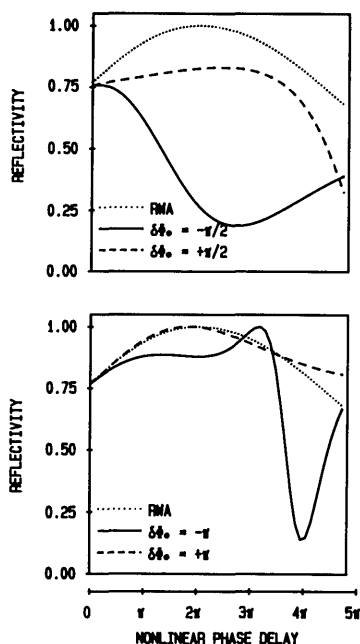


Fig. 2. Nonlinear reflection as a function of total intracavity nonlinear phase delay for a weakly linearly birefringent fiber. $\delta\Phi_0 > 0$ and $\delta\Phi_0 < 0$ correspond to input coupling close to the slow and the fast axes, respectively.

where $\beta_{x,y}$ are the propagation constants along the x and y axes, k is the free-space propagation constant, n_2 is the nonlinear refractive index, and L is the fiber length. After one round-trip an intensity-dependent nonlinear reflection coefficient R_{nl} results, which is given by

$$R_{nl} = \cos^4 \alpha + \sin^4 \alpha + 2 \cos^2 \alpha \sin^2 \alpha \cos \delta\Phi, \quad (2)$$

where $\delta\Phi = \Phi_x - \Phi_y$. The result describes a nonlinear cavity mirror, which could equally be understood in terms of an additive-pulse mode-locking mechanism³ between the two polarization eigenmodes of the fiber. An arbitrary linear phase bias between A_x and A_y may then be induced by simply controlling $\beta_{x,y}$ in the fiber. At a phase bias of $-\pi/2$ an elliptical polarization state is obtained after one round trip in the fiber, where the ellipse axes are aligned with the linearly polarized eigenmodes of the fiber and the ratio of the minor to major axis length is given by $b/a = \tan \alpha$. At this phase bias the most effective pulse shaping is obtained for $\alpha = 27.4^\circ$, where $dR_{nl}/dI = \text{maximum}$ and I is the total intensity propagating in the fiber. The comparison of Eq. (2) with reflection coefficients obtained from solutions that include nonlinear coupling¹⁵ is shown in Fig. 2. In this a phase bias of $-\pi/2$ and $\alpha = 22.5^\circ$ are assumed for a range of linear birefringences in the fiber, where $\delta\Phi_0 = (\beta_x - \beta_y)2L$. Note that to make up the phase bias of $-\pi/2$ in reflection, an additional appropriate phase delay is assumed to occur at fiber end 1 [Fig. 1(b)]. It may be seen that it is advantageous to couple close to the slow axis of the fiber to avoid the onset of nonlinear polarization instability.¹⁵ In general, when the linear birefringence is small, nonlinear nonreciprocal ellipse rotation becomes dominant and causes an early saturation of the nonlinear reflection coefficient, which is deleterious for stable

mode locking. For the general case of an elliptically birefringent fiber (a linear twist may be neglected in a coordinate system rotating with the fiber¹³) analytic expressions for a nonlinear reflection coefficient may equally be found under the RWA, since in this case the elliptically birefringent eigenmodes uncouple¹⁴ and the nonlinear phase delays are again simply given by adding the cross-phase and self-phase modulation terms.

We used an inhomogeneously broadened silica fiber doped with 1700 parts in 10^6 Nd^{3+} and emitting at $1.064 \mu\text{m}$ in the experiments. The core diameter was $5 \mu\text{m}$, and the cutoff wavelength was $1 \mu\text{m}$. We employed fiber lengths of 24 and 44 cm. A straight fiber piece had a birefringence beat length of $\approx 1 \text{ m}$ at $1.064 \mu\text{m}$. The positive GVD was $d^2\beta/d\omega^2 = 27 \text{ ps}^2/\text{km}$. Intracavity chirp compensation was achieved by using six identical prisms in series [Fig. 1(b)]. The arrangement may be described as an effective prism pair that provides an approximately ninefold increase in negative GVD compared with a standard prism pair.¹⁷ The round-trip loss of this (polarizing) arrangement was $<10\%$ using prisms (Schott SF-10) cut near the Brewster angle. The eigenmode axes and the linear phase bias in the fiber were adjusted conventionally by pressing the fiber with three piezoelectrically controlled cylinders.

With a 20% output coupler the dispersion-compensated Fabry-Perot laser cavity had a lasing threshold of 5 mW of absorbed pump power at 752 nm. The slope efficiency was 25%, and a maximum cw lasing output of 60 mW was obtained with 250 mW of absorbed pump power. The maximum cw mode-locked power was 50 mW. However, the output power was reduced by a factor of 2 in the shorter fiber length owing to limited pump-light absorption. Mode locking was initiated with an acousto-optic modulator that provided cavity-loss modulation at the round-trip frequency, which was between 40 and 80 MHz. The polarization state of the light pulses was measured in transmission at mirror M2 and in reflection by means of an optional intracavity beam splitter inserted in front of the left fiber end (Fig. 1). Pulse widths were measured at both fiber ends and at mirror M1.

With the correct polarization setting in the fiber the pulses were self-sustaining, i.e., the modulator could be switched off once pulses were obtained, and the pulses remained stable in the cavity for hours without additional adjustments, even for the shortest pulses generated. The minimum modulation required to initiate the passive pulse-forming process depended on the fiber length, the amount of overall GVD, and the lasing power. Optimum starting conditions were obtained when the nonlinearity was maximized and an amount of GVD leading to chirp-free pulses was provided. For a fiber length of 44 cm a minimum intracavity peak pulse power of 300 W was required for self-sustained operation, whereas at a lasing power of 10 mW a modulation index^{1,2} of <1 was needed to start up the passive pulse-forming process. The maximum tolerance on α for self-sustained operation was $20\text{--}35^\circ$, measured for a lasing power of 50 mW. As expected, mode-locked operation was accompanied with a reduction in b/a typically by as much as 10% and a

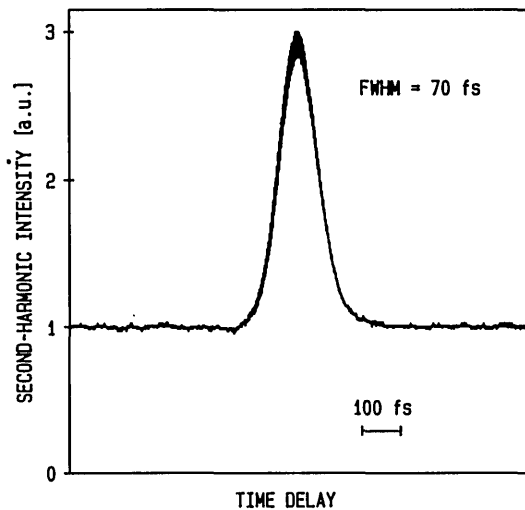


Fig. 3. Autocorrelation trace of a typical pulse generated with a 24-cm length of fiber. The pulse width is 70 fs assuming a sech^2 shape.

polarization ellipse rotation by as much as 20° (both measured at fiber end 1) compared with the free-running case. This is a further manifestation of nonlinear polarization evolution in the fiber. The intracavity pulse shaping was large, and a pulse compression by as much as a factor of 8 was measured between the pulses emerging from fiber end 1 and mirror M1. Maximum pulse compression also produced the shortest pulses, which were typically obtained for overall values of GVD between 0 and -0.3 times the fiber GVD. The autocorrelation traces of these pulses were equivalent to sech^2 pulse shapes. For the 44-cm fiber length we obtained a maximum pulse energy of 1 nJ at a repetition rate of 50 MHz with a peak output power of 4 kW. The minimum pulse lengths (assuming sech^2 pulse shapes) were 100 and 70 fs for the long and short fiber lengths, respectively. The autocorrelation trace is shown in Fig. 3. However, 70 fs was at the resolution limit of the measurement system. Indeed symmetric pulse spectra with FWHM widths of as much as 70 nm (≈ 2.3 times the inhomogeneous linewidth of the gain medium) were measured for this pulse width, which indicates that pulses shorter than 20 fs could potentially be generated by further appropriate compensation of the spatial chirp at the output mirror and second- and third-order dispersion in the cavity. Note that Raman effects were not observed in the laser, which we attribute to group-velocity walk-off between the Raman and laser signals. Finally, nonlinear polarization evolution in lasers should also permit self-starting passive mode locking. Self-starting passive mode locking was impaired in our case by the large free-running lasing bandwidth⁸ of the fiber laser of 20 nm, which was due to the combined action of

inhomogeneous broadening and spatial hole burning. A 2-order-of-magnitude reduction in the free-running lasing bandwidth could be achieved by using a homogeneously broadened Nd-phosphate glass as the fiber material.

In conclusion we have demonstrated and explained that combined passive AM and FM mode locking may be obtained by using self-phase modulation and cross-phase modulation in a single Fabry-Perot cavity. The mode-locking technique described is capable of producing pulses with a bandwidth larger than the inhomogeneous linewidth of lasers.

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